

Scaling Static Code Analysis Adoption at WhatsApp iOS

Ákos Hajdu, Jorge Mendez, Sander van Valkenburg, Artem Kupriianets,
Matteo Marescotti, Dulma Churchill, Sopot Cela

Meta



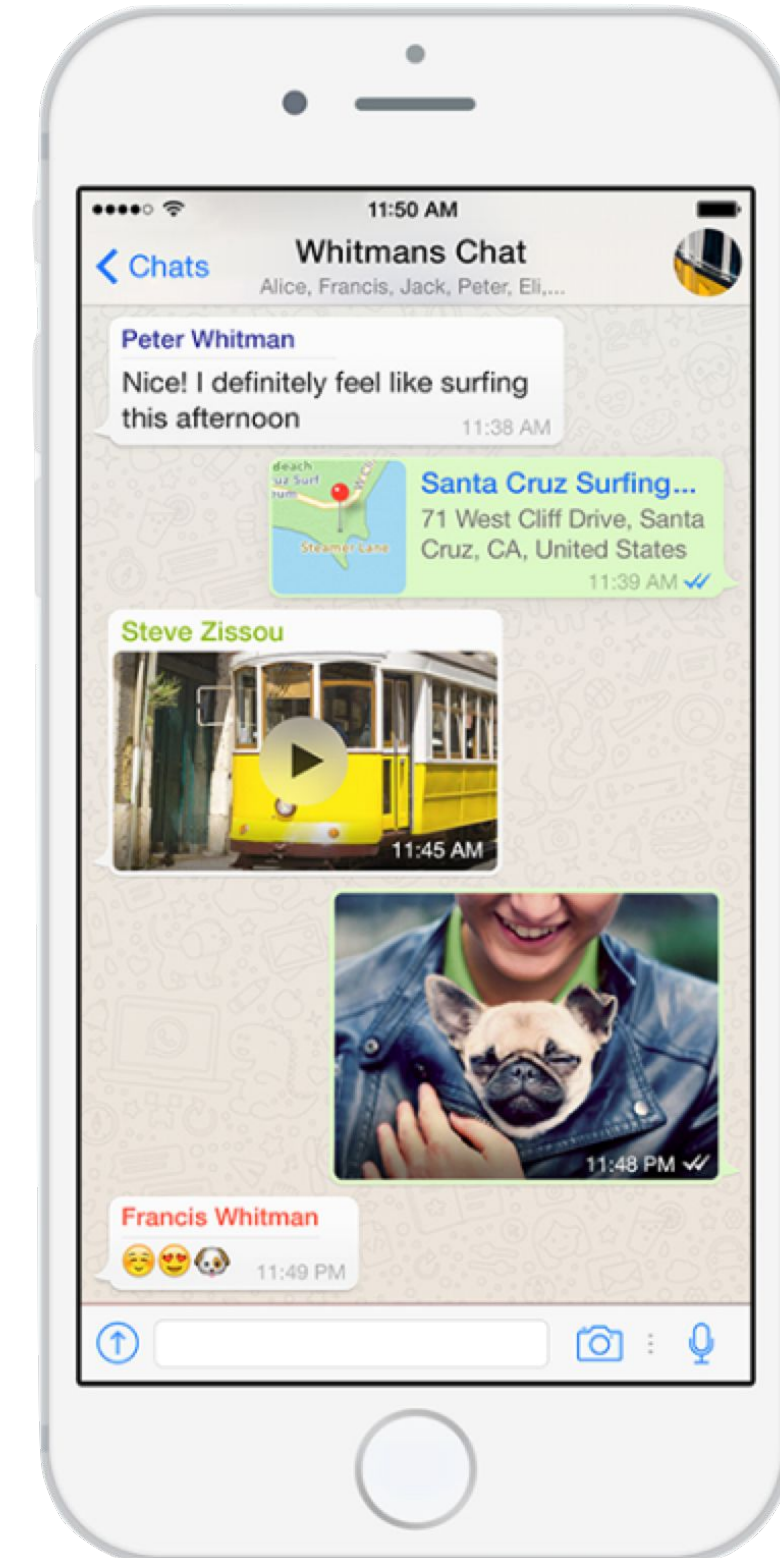
Goal

Bring **static analysis** to WhatsApp iOS

Focus on **memory management** problems

Prevent regressions

Clean up pre-existing issues



Infer



Open-source static analysis platform

Obj-C (and Swift) frontends

Checkers

Block parameter not null checked

Captured strong self

Mixed self-weakself

Nil insertion into collection

Self in block passed to init

Strong self not checked

```
- (void)doTask:(void (^)(void))completion {  
    // Do the task ...  
    completion();  
}
```

```
__weak __typeof(self) weakSelf = self;  
int (^my_block)() = ^() {  
    __strong __typeof(weakSelf) strongSelf = weakSelf;  
    if (strongSelf) {  
        [strongSelf foo];  
        int x = self->x; // typo here  
    }  
    return 0;  
};
```

Clang Static Analyzer



Built on Clang + LLVM

Deployed via clang-tidy

Focus on nullability

- Null(able) passed to nonnull

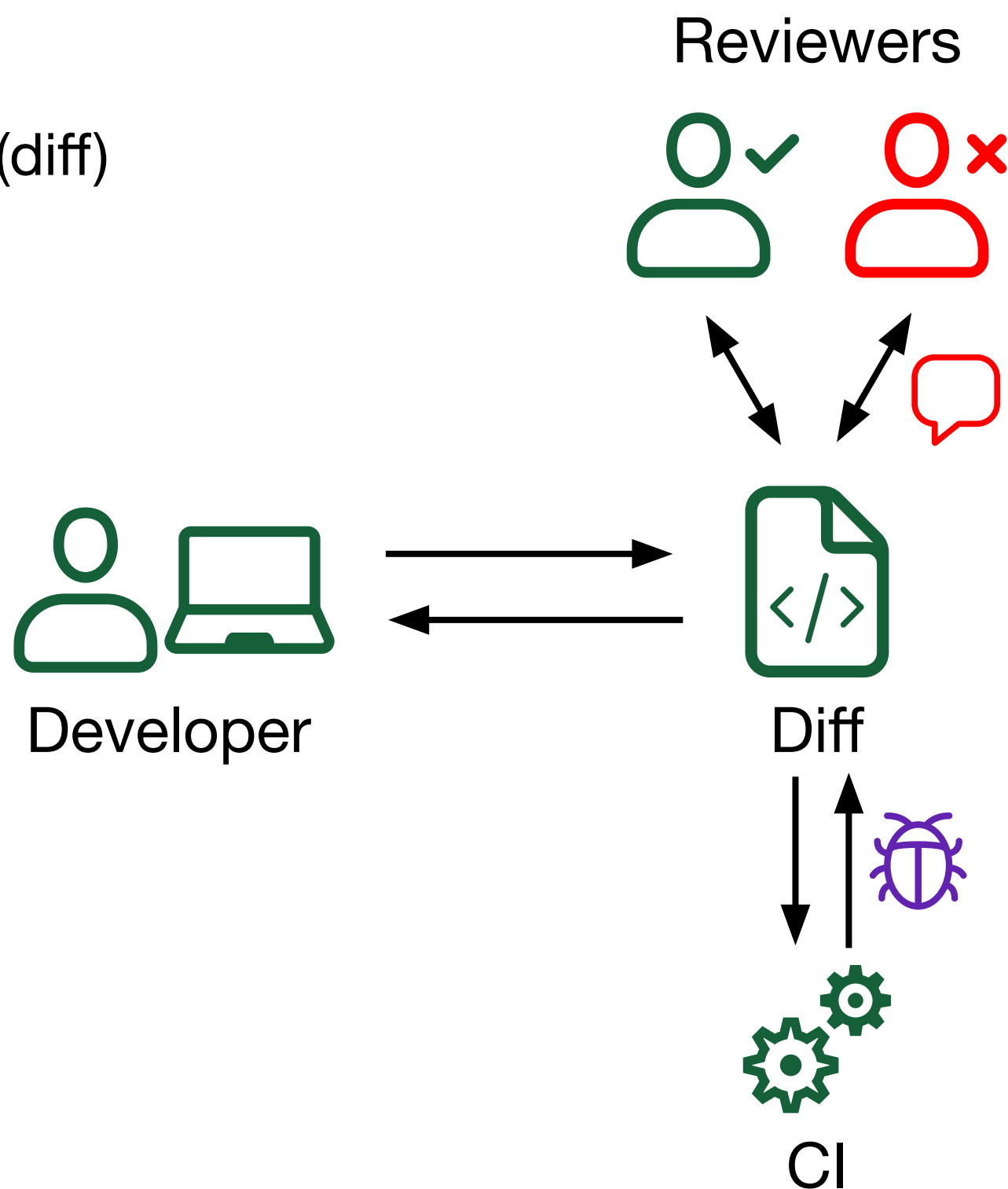
- Null(able) returned from nonnull

- Nullable dereferenced

```
- (nonnull NSString *)firstChar:(NSString *)str {  
    if (str.length == 0) {  
        return nil;  
    }  
    return [str substringToIndex:1];  
}
```

Deployment

Code changes (diff)



```
7 - (void)doTask:(void (^)(void))completion {
8   // Do the task ...
9   completion();
```

✗ Block Parameter Not Null
Checked(BLOCK_PARAMETER_NOT_NULL_CHECKED) raised an error on **line 9**

The block `completion` is executed without a check for nil at line 9, column 5. This could cause a crash. Docs: https://fbinfer.com/docs/next/all-issue-types#block_parameter_not_null_checked

```
12 - (nonnull NSString *)firstChar:(NSString *)str {
13   if (str.length == 0) {
14     return nil;
```

⚠ clang-analyzer-nullability.NullReturnedFromNonnull raised a warning on **line 14**

nil returned from a method that is expected to return a non-null value
WATestAnalyzerWarnings/Source/WATestAnalyzerWarnings.m:13:9: note: Assuming the condition is true

```
if (str.length == 0) {
  ^
```

WATestAnalyzerWarnings/Source/WATestAnalyzerWarnings.m:13:5: note: Taking true branch

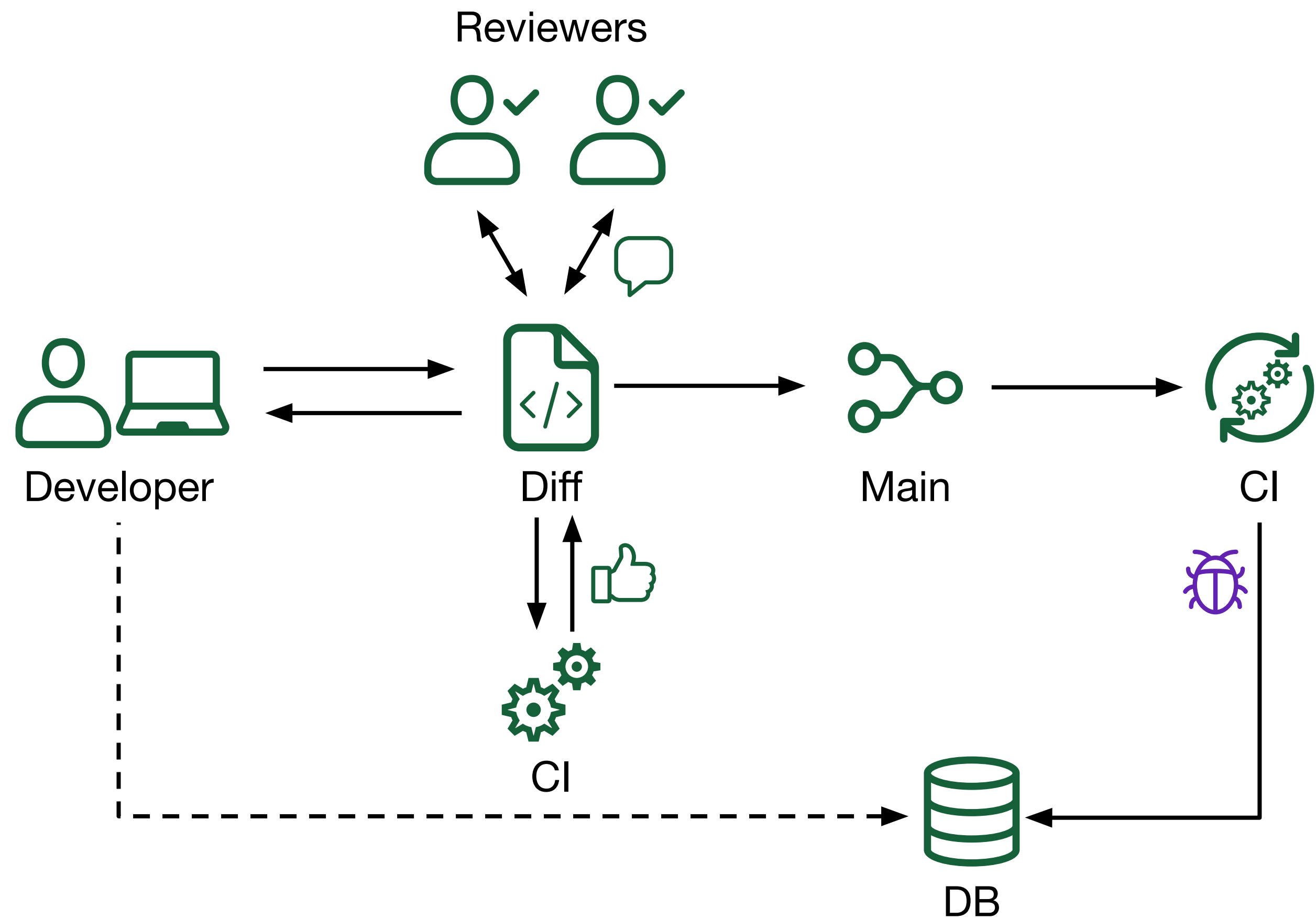
```
if (str.length == 0) {
  ^
```

WATestAnalyzerWarnings/Source/WATestAnalyzerWarnings.m:14:9: note: nil returned from a method that is expected to return a non-null value

```
return nil;
  ^
```


Deployment

Continuous



Analyzing code changes



821

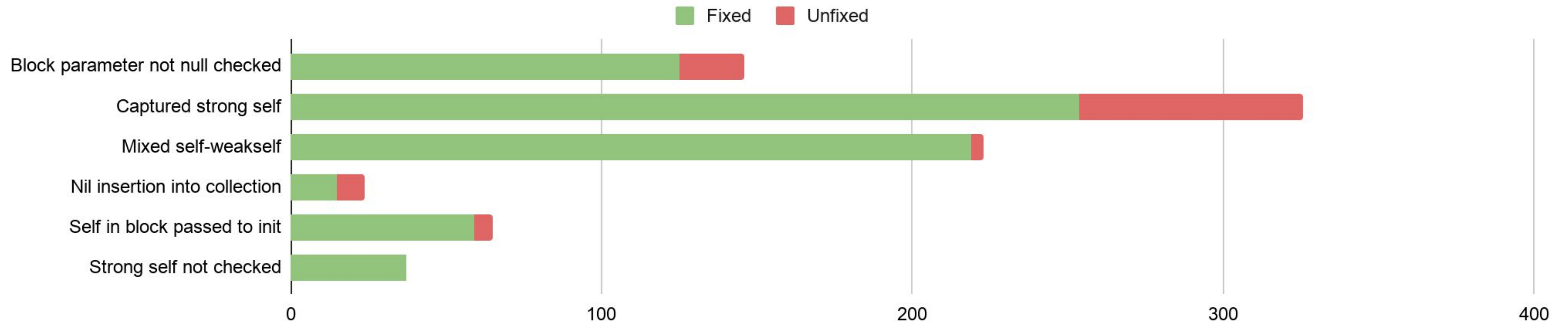
Reported (1y)

709

Fixed

86%

Fixrate



Worked out-of-the-box: targeted checkers, perfected on other deployments

Clang Static Analyzer



2419

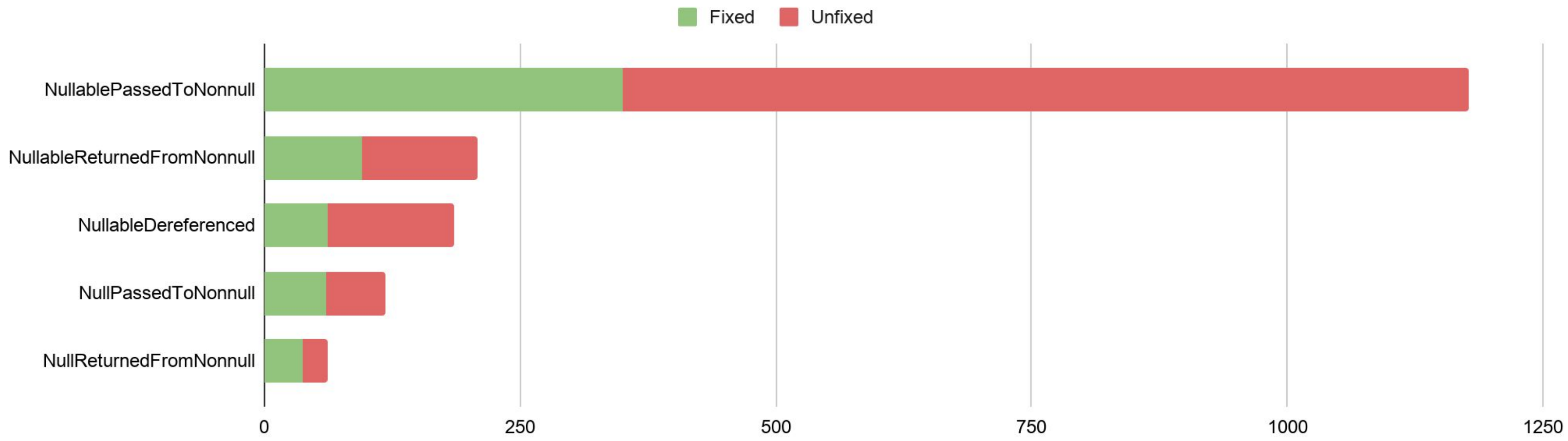
Reported (9mo)

1005

Fixed

42%

Fixrate



Clang Static Analyzer



Conservative about **side effects**

```
if (obj.prop) {  
    expectsNonNull(obj.prop);  
}
```

Pre-existing issues

Changed line-based filtering

```
33 }  
34 -- (void)takesNonNull:(NSString* _Nonnull)arg {  
34 +- (void)takesNonNull:(NSString* _Nonnull)arg count:(int)count {  
35 }  
36 - (void)callWithNull_Bad {  
37     NSString* s = nil;  
38 - [self takesNonNull:s];  
38 + [self takesNonNull:s count:0];  
39 }
```

Base revision relint & diff → 70% fixrate

Pre-existing issue cleanup

Human fixes

Power users

- Passionate about code quality
- Feedback on early prototypes
- Help with advocating



50+		
Engineers mobilized		
664	1186	
Fixed	Total	
144	739	
Fixed	Total	

Human fixes

UX matters

Internal database + basic UI

Intuitive UI with task tracking



50+	
Engineers mobilized	
664	1186
Fixed	Total
144	739
Fixed	Total

Human fixes

Being present

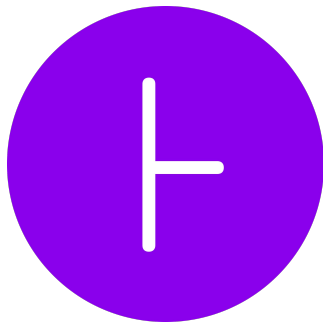
50+

Engineers mobilized

Observe signal

Do first assessment

Listen to feedback



664

Fixed

1186

Total



144

Fixed

739

Total

Human fixes

Gamification

Leaderboard

Team-up events

In-person/remote get together

Advertised on internal platform



50+

Engineers mobilized

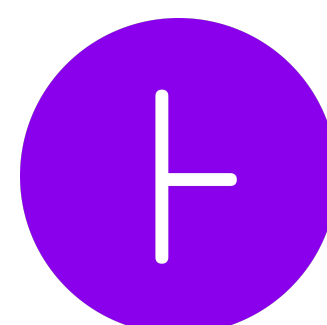
664 1186

Fixed Total

144 739

Fixed Total

GenAI fixes



175
Changes

90
Accepted

115
Issues fixed



121
Changes

76
Accepted

Bugs slipped through
False positive hallucinations
Misaligned incentives

Mitigations
Initial assessment
Upfront alignment

Review is still bottleneck

Conclusions

Infer + Clang Static Analyzer for WhatsApp iOS

Code changes + pre-existing issues

Human fixes

- Power users

- UX matters

- Being present

- Gamification

- Team up events

GenAI

- Initial assessment or upfront alignment

- Review still bottleneck



1700+

Regressions prevented

1000+

Issues cleaned up

50+

Engineers mobilized